



“The desire to fly is an idea handed down to us by our ancestors who... looked enviously on the birds soaring freely through space... on the infinite highway of the air...”

WILBUR WRIGHT

THE PREHISTORY OF FLIGHT

THE PATH TO POWERED FLIGHT WAS OPENED UP BY DREAMERS AND ODDBALL INVENTORS WHO BRAVED BOTH PUBLIC RIDICULE AND PHYSICAL INJURY

HUMAN BEINGS HAVE always dreamed of flight. They did not, however, dream of the Boeing 747. The flight to which humans traditionally aspired was that of the birds, a business of feathers and flapping wings. To this the myths and legends of many cultures testify. In the most famous of these ancient stories, the skilled craftsman Daedalus makes wings of feathers and wax so he and his son Icarus can escape their imprisonment on the island of Crete. The technology improbably works, but Icarus flies too close to the sun and melts the wax, falling to his doom.

The illusion that a person could fly like a bird or a bat cost some brave and foolish men their lives or limbs. The historical record is scattered with “tower jumpers” who launched themselves into the air supported only by blind faith and poorly improvised wings. In 1178, for example, in Constantinople, a follower of Islam chose the moment of a visit to the Christian Byzantine Emperor by a Muslim sultan to demonstrate his powers of flight, jumping off a high building in a copious white robe stiffened with willow sticks. In the words of a later flight experimenter, Octave Chanute, “the weight of his body having more power to draw him downward than his artificial

wings had to sustain him, he fell and broke his bones.” Other recorded attempts – by the learned Moor Abbas ibn-Firnas in Andalusia in 875, by English monk Oliver of Malmesbury in the 11th century, by Giovanni Battista Danti in Perugia, Italy, in 1499 – all had the same result for the same reason.

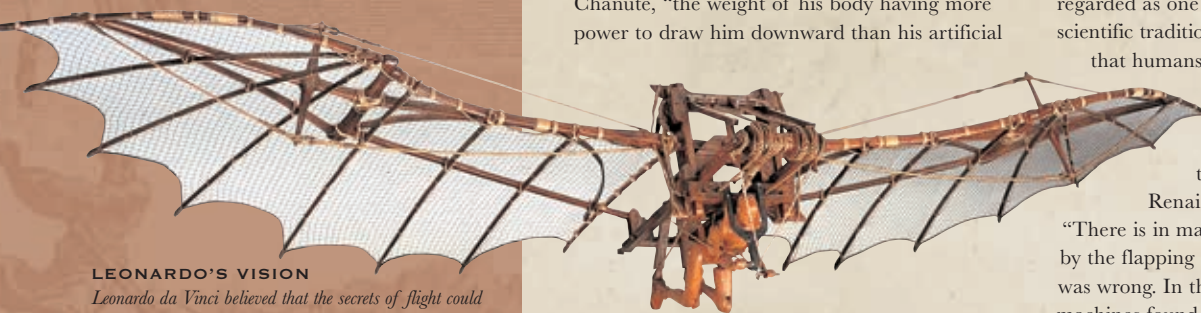
“Instruments to fly”

Myth and folklore were also rich in tales of airworthy vehicles that might carry the weight of a human, from various “chariots of the gods” to witches’ broomsticks. The idea of a “flying machine” was picked up by English philosopher-monk Roger Bacon in the 13th century – a man regarded as one of the founders of the modern scientific tradition. Bacon declared himself certain that humans could build “instruments to fly,”

involving a mechanism that would flap wings. Such “ornithopters” also obsessed the imagination of Italian

Renaissance genius Leonardo da Vinci.

“There is in man [the ability] to sustain himself by the flapping of wings,” Leonardo wrote. He was wrong. In the many sketches of flying machines found in Leonardo’s notebooks, the only truly promising idea is a screwlike propeller that he hoped would spiral into the air – a remote



LEONARDO'S VISION

Leonardo da Vinci believed that the secrets of flight could be learned by studying birds. His concept of a flying machine (model shown here) was as impractical as all other devices for muscle-powered, flapping-wing flight.